

Object Oriented Analysis and Design

Mini Project Submission – Phase 2

Project Title: Research and Collaboration System

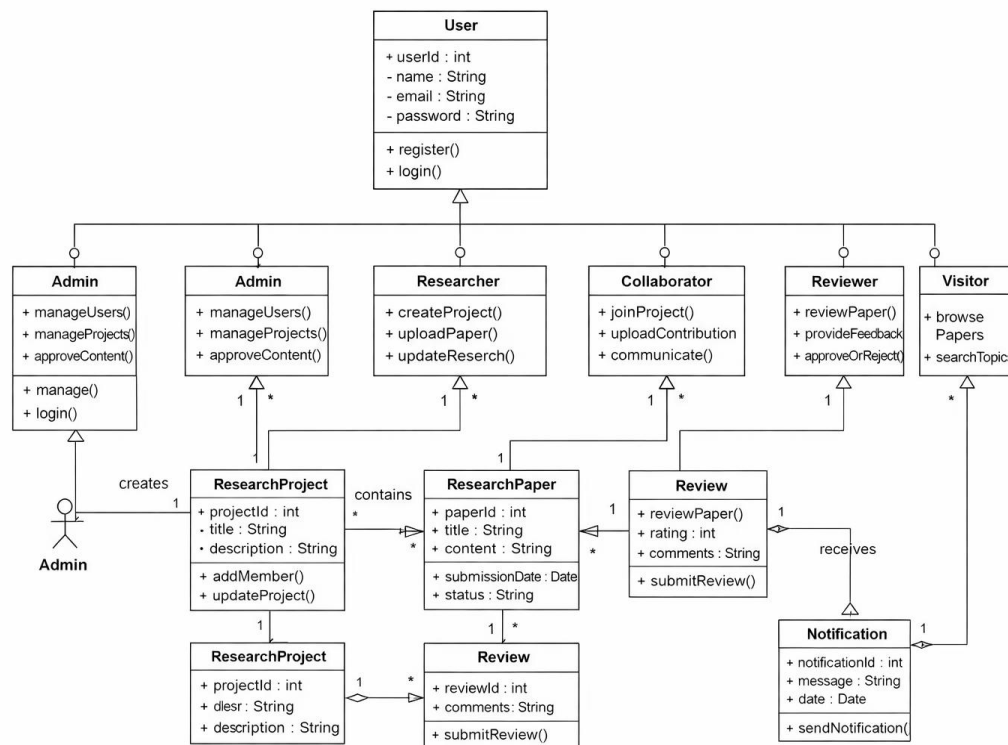
Team Number: Team-15

Team Members:

1. Salar Hafeez – PES1UG24CS827
2. M Revanth Reddy – PES1UG23CS344
3. Magalur Adarsh – PES1UG23CS338

Phase-2: Class Diagram

Research and Collaboration System - Class Diagram



Brief Description of the System:

The Research and Collaboration System is an education-based platform designed to support structured research activities. The system allows researchers to create research projects, upload papers, collaborate with others, and receive reviews. It follows object-oriented principles to represent system entities and their relationships clearly using a class diagram.

Description of the Class Diagram:

The class diagram represents the static structure of the Research and Collaboration System. It identifies the main classes, their attributes, methods, and the relationships between them. The design is based on role-based access and responsibilities defined in Phase-1 use case diagram.

Classes Identified:

User:

The User class is the base class containing common attributes such as userId, name, email, and password. It provides basic functionalities like register and login.

Admin:

The Admin class inherits from the User class and is responsible for managing users, managing research projects, and approving or removing content.

Researcher:

The Researcher class represents users who create research projects, upload research papers, and update research work.

Collaborator:

The Collaborator class allows users to join research projects, upload contributions, and communicate with researchers.

Reviewer:

The Reviewer class is responsible for reviewing submitted research papers, providing feedback, and approving or rejecting papers.

Visitor:

The Visitor class allows general users to browse and search publicly available research papers without modifying any content.

ResearchProject:

This class represents a research project. It contains attributes like projectId, title, description, and status. It supports adding members and updating project details.

ResearchPaper:

The ResearchPaper class stores information related to research papers such as paperId, title, content, submission date, and status.

Review:

The Review class captures reviewer feedback including rating and comments for a research paper.

Notification:

The Notification class is used to send alerts and updates to users regarding system activities.

Conclusion:

The class diagram clearly defines the structure of the Research and Collaboration System. It shows inheritance, associations, and responsibilities of each class. This design ensures modularity, scalability, and easy maintenance of the system.